

# Björn Müller

## Curriculum Vitae

✉ [muellerb@mps.mpg.de](mailto:muellerb@mps.mpg.de)

### Research Interests

Inverse problems for wave equations, passive imaging, regularization theory, computational inverse problems, wave propagation, and helioseismology.

### Academic Positions

- since 12/2023 **Postdoctoral Researcher**, *Max-Planck-Institute for Solar System Research*, Supervisor: Prof. Dr. Laurent Gizon  
Research on inverse problems and passive imaging for wave equations with applications to helioseismology; uniqueness and stability theory for inverse parameter and source problems; iterative regularization methods and numerical implementations.
- 2021–2025 **Associated Member**, *Collaborative Research Center 1456*  
Project C04: Correlations of solar oscillations: modeling and inversions
- 02/2019 – 07/2019 **Research Assistant**, *Institute of Numerical and Applied Mathematics, Georg-August-University Göttingen*, Supervisor: Prof. Dr. Thorsten Hohage  
Development of a Python toolbox for iterative regularization

### Education

- 04/2020 – 11/2023 **Ph.D. Physics, summa cum laude**, *GAUSS Program, Georg-August-University Göttingen, Iterative helioseismic holography*  
Supervisors: Prof. Dr. Laurent Gizon, Prof. Dr. Thorsten Hohage, Dr. Damien Fournier
- 03/2019 – 03/2020 **M.Sc. Physics, with distinction**, *Georg-August-University Göttingen, Reverberation Mapping of Active Galactic Nuclei*  
Final grade: 1.0; Supervisors: Prof. Dr. Wolfram Kollatschny, Martin Ochmann
- 10/2015 – 02/2019 **B.Sc. Mathematics, with distinction**, *Georg-August-University Göttingen, Inversion of Meridional Flow Using Pinsker Estimator*  
Final grade: 1.2; Supervisors: Prof. Dr. Laurent Gizon, Prof. Dr. Thorsten Hohage, Dr. Damien Fournier, Dr. Vincent Böning
- 10/2015 – 02/2019 **B.Sc. Physics**, *Georg-August-University Göttingen, Inversion of Meridional Flow Using Pinsker Estimator*  
Final grade: 1.2; Supervisors: Prof. Dr. Laurent Gizon, Prof. Dr. Thorsten Hohage, Dr. Damien Fournier, Dr. Vincent Böning

### Prizes and Awards

- 2026 **IOP Trusted Reviewer status**
- 2025 **IOP Best Paper Award**, *Inverse Problems*  
For the paper “Quantitative passive imaging by iterative holography: the example of helioseismic holography”
- 2025 **Otto Hahn Medal**, *Max Planck Society*  
For the work on “iterative helioseismic holography”

### Peer-Reviewed Publications

**Quantitative passive imaging by iterative holography: the example of helioseismic holography**, B. Müller, T. Hohage, L. Gizon and D. Fournier, *Inverse Problems* (2024), Volume 40(4), 045016, [link](#)

**The Great Slump: Mrk 926 reveals discrete and varying Balmer line satellite components during a drastic phase of decline**, W. Kollatschny, M. W. Ochmann, S. Kaspi, C. Schumacher, E. Behar, D. Chelouche, K. Horne, B. Müller, S. E. Rafter, R. Chini, M. Haas and M.A. Probst, *A&A* (2022), Volume 657, A122, [link](#)

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## Preprints

**Holographic X-ray Phase Contrast Imaging with Partial Coherence: Uniqueness and Reconstructions from Intensity Correlations**, T. Hohage, M. Karimi, B. Müller, arXiv preprint (2026), [link](#)

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## Conference Proceedings

**Uniqueness results in time-harmonic passive imaging with applications to helioseismology**, T. Hohage, B. Müller, 17th International Conference on Mathematical and Numerical Aspects of Wave Propagation (2026), Montreal, Canada, [link](#), p.165

**Stability of a passive inverse parameter problem with applications to helioseismology**, B. Müller, T. Hohage, 17th International Conference on Mathematical and Numerical Aspects of Wave Propagation (2026), Montreal, Canada, [link](#), p.245

**On quantitative passive imaging**, T. Hohage, B. Müller, D. Fournier and L. Gizon, 16th International Conference on Mathematical and Numerical Aspects of Wave Propagation (2024), Berlin, Germany, [link](#), p.197

**Using iterative helioseismic holography for inversions of solar differential rotation**, B. Müller, T. Hohage, L. Gizon and D. Fournier, 16th International Conference on Mathematical and Numerical Aspects of Wave Propagation (2024), Berlin, Germany, [link](#), p.211

**Iterative helioseismic holography-Inversions for solar differential rotation**, B. Müller, L. Gizon, T. Hohage and D. Fournier, 15th International Conference on Mathematical and Numerical Aspects of Wave Propagation (2022), Paris, France, [link](#), p.314

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## Selected Manuscripts in Preparation

**Unique identifiability of interior parameters in passive imaging with applications to helioseismology**, B. Müller and T. Hohage

**Stability of the inverse parameter problem in passive imaging with applications to helioseismology**, B. Müller and T. Hohage

**Nonuniqueness of simultaneous reconstruction of potentials and source strength in passive imaging**, B. Müller and T. Hohage

**Unique identifiability of finite-measure source powers from passive near-field measurements**, B. Müller and T. Hohage

**Iterative helioseismic holography of solar differential rotation applied to synthetic data**, B. Müller, L. Gizon, D. Fournier and T. Hohage

**Improved helioseismic holography of solar meridional flows applied to synthetic data**, B. Müller, L. Gizon, D. Fournier and T. Hohage

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## Research Software

- 2019– **RegPy: Regularization tools for inverse and ill-posed problems in Python**, *Main developer*  
[project page](#). Developed numerical methods and software for regularization and inverse problems.
- 2019– **Gecho toolbox**, *Main developer*  
To be published. Spectroscopic time-series analysis

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## Selected Conferences and Talks

- 22.06.2026– **17th International Conference on Mathematical and Numerical Aspects of Wave Propagation (WAVES2026)**, *Montreal, Canada*  
26.06.2026 Talk: *Stability of a passive inverse parameter problem with applications to helioseismology*
- 31.05.2026– **12th International Conference: “Inverse Problems: Modelling and Observations”**, *Malta*  
05.06.2026 Talk: *Unique identifiability of interior parameters in passive imaging with applications to helioseismology*
- 24.06.2025– **Annual Meeting of the Max Planck Society**, *Magdeburg, Germany*  
26.06.2025 Talk: *Iterative helioseismic holography*
- 17.02.2025– **SDO 2025 Science Workshop**, *Boulder, USA*  
21.02.2025 Talk: *Measuring the solar differential rotation with iterative helioseismic holography*
- 21.10.2024– **The Whole Sun 2024 Scientific Meeting**, *San Cristóbal de La Laguna, Spain*  
25.10.2024 Talk: *Quantitative iterative helioseismic holography*
- 30.06.2024– **16th International Conference on Mathematical and Numerical Aspects of Wave Propagation (WAVES2024)**, *Berlin, Germany*  
05.07.2024 Talk: *Using iterative helioseismic holography for inversions of solar differential rotation*
- 26.05.2024– **11th International Conference: “Inverse Problems: Modelling and Observations”**, *Malta*  
01.06.2024 Talk: *Unique identifiability of a passive inverse parameter problem: The example of helioseismology*
- 04.09.2023– **11th Applied Inverse Problems Conference**, *Göttingen, Germany*  
08.09.2023 Talk: *Quantitative passive imaging by iterated helioseismic holography*
- 25.09.2023– **6th European Conference on Computational Optimization**, *Heidelberg, Germany*  
27.09.2023 Talk: *Quantitative passive imaging by iterated helioseismic holography*
- 01.03.2023 **The Good Vibrations Seminar**, *Online*  
Talk: *Validating iterative helioseismic holography on synthetics for differential rotation and meridional flows*

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## Teaching

- 04/2017 – **Tutor for various lectures**, *Mathematical Institute and Physics Institute, Georg-August-University Göttingen*  
09/2022 Foundations of Stochastics; Statistics in Geosciences; Experimental Physics 4; Advanced Lab Course; Physics Laboratory Course for Minor Subjects; Classical Field Theory

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## Professional Service

- 2026– **Reviewer for *Inverse Problems***
- 2024– **Member of the DEICO Committee (Diversity, Equity and Inclusion) at MPS**

2020–2023   **Organization of the buddy group at MPS**

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## Additional Academic Outputs

**Iterative helioseismic holography**, B. Müller, Ph.D. thesis, Georg-August-University Göttingen, [link](#)

**VizieR Online Data Catalog: Mrk 926 spectra (Kollatschny+, 2022)**, *W. Kollatschny, M. W. Ochmann, S. Kaspi, C. Schumacher, E. Behar, D. Chelouche, K. Horne, B. Müller, S. E. Rafter, R. Chini, M. Haas and M.A. Probst*, VizieR Online Data Catalog: Mrk 926 (2021), [link](#)

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## Languages

**German**, *Native*

**English**, *Fluent / professional proficiency*

**Spanish**, *Basic knowledge*